



CAPS DE LA APLICACION EN FUNCIONAMIENTO CONJUNTO A LA API

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ASIGNATURA: APLICACIONES DISTRIBUIDAS

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API EN SPRING

```
1 package com.denn.productos.modelo.entidad;
2
3 import java.io.Serializable;
4
5 import jakarta.persistence.CascadeType;
6 import jakarta.persistence.Entity;
7 import jakarta.persistence.FetchType;
8 import jakarta.persistence.GeneratedValue;
9 import jakarta.persistence.GenerationType;
10 import jakarta.persistence.Id;
11 import jakarta.persistence.JoinColumn;
12 import jakarta.persistence.ManyToOne;
13 import lombok.Data;
14
15 @Data
16 @Entity
17 public class Producto implements Serializable{
18
19     private static final long serialVersionUID = 1L;
20
21 }
```

Console

```
2025-10-27T00:10:16.739-05:00 INFO 20292 --- [productos] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet
2025-10-27T00:10:16.741-05:00 INFO 20292 --- [productos] [nio-8080-exec-1] w.s.m.s.DefaultHandlerException
2025-10-27T00:10:16.743-05:00 WARN 20292 --- [productos] [nio-8080-exec-1] w.s.m.s.DefaultHandlerException
2025-10-27T00:10:16.745-05:00 WARN 20292 --- [productos] [nio-8080-exec-1] w.s.m.s.DefaultHandlerException
2025-10-27T00:10:16.747-05:00 WARN 20292 --- [productos] [nio-8080-exec-1] w.s.m.s.DefaultHandlerException
```

POSTMAN

GET http://localhost:8080/api/producto

Send

Params Authorization Headers (0) Body Scripts Settings Cookies

Query Params

Key	Value	Description
Key	Value	Description

Body Cookies Headers (5) Test Results

200 OK - 24 ms - 834 B

JSON Preview Visualize

```
1 {
2   "idProducto": 1,
3   "nombreProducto": "laptop lenovo ",
4   "descripcionProducto": "hh",
5   "stockProducto": 0,
6   "precioProducto": 700.0,
7   "estadoProducto": "Activo",
8   "idCategoria": 1,
9   "nombreCategoria": "Libros",
10  "estado": "Activo"
11 }
```

VISUAL STUDIO 2022

ClientesLINQO

```
1 using Microsoft.AspNetCore.Mvc;
2 using Microsoft.EntityFrameworkCore;
3 using System.Linq;
4 using System.Net.Http;
5 using System.Text.Json;
6 using System.Threading.Tasks;
7 using Microsoft.Extensions.Logging;
8 using Microsoft.Extensions.DependencyInjection;
9 using Microsoft.Extensions.Hosting;
10 using Microsoft.Extensions.Configuration;
11 using Microsoft.Extensions.Options;
12 using Microsoft.Extensions.DependencyInjection.Extensions;
13 using Microsoft.Extensions.Logging.Abstractions;
14 using Microsoft.Extensions.Options.DefaultImplementation;
15 using Microsoft.Extensions.Options.DefaultImplementation;
16 using Microsoft.Extensions.Options.DefaultImplementation;
17 using Microsoft.Extensions.Options.DefaultImplementation;
18 using Microsoft.Extensions.Options.DefaultImplementation;
19 using Microsoft.Extensions.Options.DefaultImplementation;
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96 using Microsoft.Extensions.Options.DefaultImplementation;
97 using Microsoft.Extensions.Options.DefaultImplementation;
98 using Microsoft.Extensions.Options.DefaultImplementation;
99 using Microsoft.Extensions.Options.DefaultImplementation;
100 using Microsoft.Extensions.Options.DefaultImplementation;
```

Console

```
info: System.Net.Http.HttpClient.IProductService.LogicalHandler[100]
      Start processing HTTP request GET http://localhost:8080/api/producto
info: System.Net.Http.HttpClient.IProductService.LogicalHandler[100]
      Sending HTTP request GET http://localhost:8080/api/producto
info: System.Net.Http.HttpClient.IProductService.LogicalHandler[101]
      Received HTTP response headers after 8.3947ms - 200
info: System.Net.Http.HttpClient.IProductService.LogicalHandler[101]
      End processing HTTP request after 21.2633ms - 200
info: System.Net.Http.HttpClient.IProductService.LogicalHandler[100]
      Start processing HTTP request POST http://localhost:8080/api/producto
info: System.Net.Http.HttpClient.IProductService.LogicalHandler[100]
      Sending HTTP request POST http://localhost:8080/api/producto
info: System.Net.Http.HttpClient.IProductService.LogicalHandler[101]
      Received HTTP response headers after 8.1124ms - 200
info: System.Net.Http.HttpClient.IProductService.LogicalHandler[101]
      End processing HTTP request after 96.6987ms - 200
info: System.Net.Http.HttpClient.IProductService.LogicalHandler[100]
      Start processing HTTP request GET http://localhost:8080/api/producto
info: System.Net.Http.HttpClient.IProductService.LogicalHandler[100]
      Sending HTTP request GET http://localhost:8080/api/producto
info: System.Net.Http.HttpClient.IProductService.LogicalHandler[101]
      Received HTTP response headers after 10.3425ms - 200
info: System.Net.Http.HttpClient.IProductService.LogicalHandler[101]
      End processing HTTP request after 17.2153ms - 200
```



MODULOS

Lista de Productos - Producto x +

localhost:7032/Products

ProductosMvcClient Home Privacy

Lista de Productos

[Nuevo Producto](#)

ID	Nombre	Descripción	Precio	Stock	Estado	Categoría ID	Acciones
1	laptop lenovo	hh	\$700,00	0	Activo	5	Editar Eliminar
2	refrigeradora	lg	\$900,00	0	Activo	8	Editar Eliminar
3	Rayuela-Cortazar	ppp	\$400,00	0	Activo	5	Editar Eliminar

Crear Producto - ProductoMvcClient x +

localhost:7032/Products/Create

ProductosMvcClient Home Privacy

Crear Nuevo Producto

Nombre del Producto
MICROFONO

Descripción
CARO

Precio
1000

Categoría
Electrónicos

[Guardar](#) [Cancelar](#)

localhost:7032/Products

ProductosMvcClient Home Privacy

Lista de Productos

[Nuevo Producto](#)

ID	Nombre	Descripción	Precio	Stock	Estado	Categoría ID	Acciones
1	laptop lenovo	hh	\$700,00	0	Activo	5	Editar Eliminar
2	refrigeradora	lg	\$900,00	0	Activo	8	Editar Eliminar
3	Rayuela-Cortazar	ppp	\$400,00	0	Activo	5	Editar Eliminar
4	MICROFONO	CARO	\$1.000,00	0	Activo	1	Editar Eliminar



POSTGRES SQL

The screenshot shows the PostgreSQL 17 interface. The left sidebar displays the database structure, including schemas, tables, and views. The main window shows a query history with the following query:

```
1 SELECT * FROM public.producto
2 ORDER BY id_producto ASC
```

The Data Output section shows the results of the query, displaying columns and their values:

id_producto	descripcion_producto	etiqueta_producto	nombre_producto	precio_producto	stock_producto	id_categoria
1	tn	Activo	lapta lenovo	700	0	5
2	tz	Activo	refrigeradora	800	0	8
3	zoo	Activo	Refrigerador	400	0	5
3	zoo	Activo	Refrigerador	400	0	5

The screenshot shows the PostgreSQL 17 interface. The left sidebar displays the database structure. The main window shows the same query as the previous screenshot:

```
1 SELECT * FROM public.producto
2 ORDER BY id_producto ASC
```

The Data Output section shows the results of the query, displaying columns and their values:

id_producto	descripcion_producto	etiqueta_producto	nombre_producto	precio_producto	stock_producto	id_categoria
1	tn	Activo	lapta lenovo	700	0	5
2	tz	Activo	refrigeradora	800	0	8
3	zoo	Activo	Refrigerador	400	0	5
4	CARD	Activo	REPRODUCIDOR	1000	0	1

Below the results, there are two green success messages:

- ✓ Successfully ran. Total query runtime: 198 msec. 4 rows affected.
- ✓ PostgreSQL 17/api.productos - Database connected.